Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Design FIR filter with windowing method.	
Experiment - 6	Date: 14/10/2026	Enrolment No:92400133050

Aim : Design FIR filter with windowing method.

1. Design FIR Low pass filter having cutoff frequency 3KHz, N=31; Sampling frequency=10KHz
2. Design FIR High pass filter having cutoff frequency 3KHz, N=31; Sampling frequency=10KHz
3. Design FIR Band pass filter having cutoff frequency(3KHz, 6KHz) N=31; Sampling frequency=10KHz
4. Design FIR Band stop filter having cutoff frequency(3KHz, 6KHz) N=31; Sampling frequency=10KHz

Code :

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import firwin, freqz

# Step 1: Filter parameters
N = 31

# Step 2: Design four FIR filters
low = firwin(N, 3000, window='hann',
             pass_zero='lowpass', fs=10000)

high = firwin(N, 3000, window='hann',
              pass_zero='highpass', fs=10000)

band = firwin(N, [1000, 6000], window='hann',
              pass_zero='bandpass', fs=13000)

reject = firwin(N, [1000, 6000], window='hann',
```

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```
        pass_zero='bandstop', fs=13000)

# Step 3: Calculate frequency responses
filters = [low, high, band, reject]
titles = ['Low-Pass Filter', 'High-Pass Filter',
        'Band-Pass Filter', 'Band-Reject Filter']

# Step 4: Plot all graphs
plt.figure(figsize=(10, 8))

for i in range(4):
    f, H = freqz(filters[i], worN=2048, fs=10000)
    magnitude_db = 20 * np.log10(np.maximum(np.abs(H), 1e-10))

    plt.subplot(2, 2, i + 1)
    plt.plot(f, magnitude_db)
    plt.title(titles[i])
    plt.xlabel('Frequency (Hz)')
    plt.ylabel('Magnitude (dB)')
    plt.grid()

plt.tight_layout()
plt.show()
# impluse response
n = np.arange(N)

plt.figure(figsize=(10, 8))

for i in range(4):
    plt.subplot(2, 2, i + 1)
    plt.stem(n, filters[i])
    plt.title(titles[i])
```

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```
    plt.xlabel('Sample (n)')
    plt.ylabel('Amplitude')
    plt.grid()

plt.tight_layout()
plt.show()
# Phase Response
Fs = 10000  # Sampling frequency
plt.figure(figsize=(10, 8))

for i in range(4):
    f, H = freqz(filters[i], worN=2048, fs=Fs)

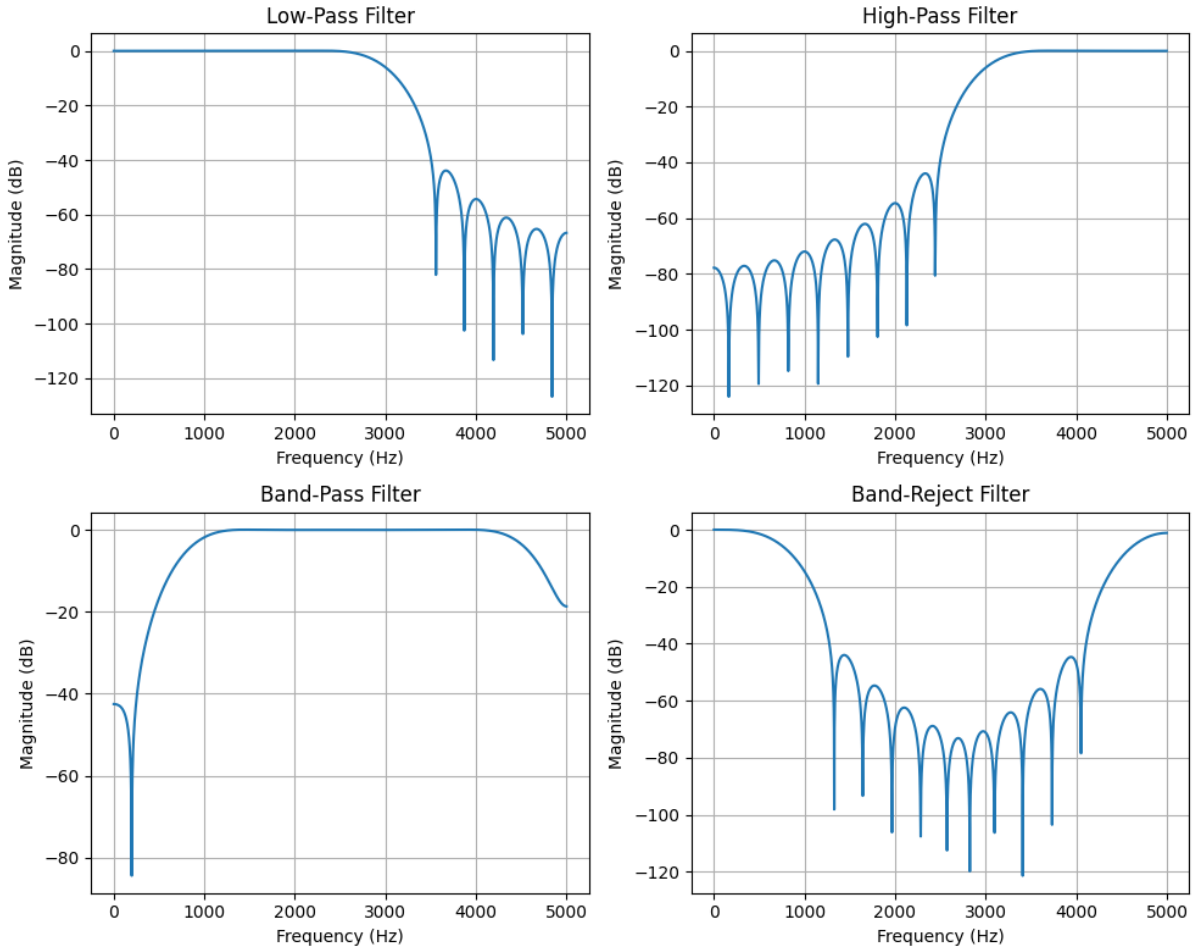
    phase = np.unwrap(np.angle(H))

    plt.subplot(2, 2, i + 1)
    plt.plot(f, phase)
    plt.title(titles[i])
    plt.xlabel('Frequency (Hz)')
    plt.ylabel('Phase (Radians)')
    plt.grid()

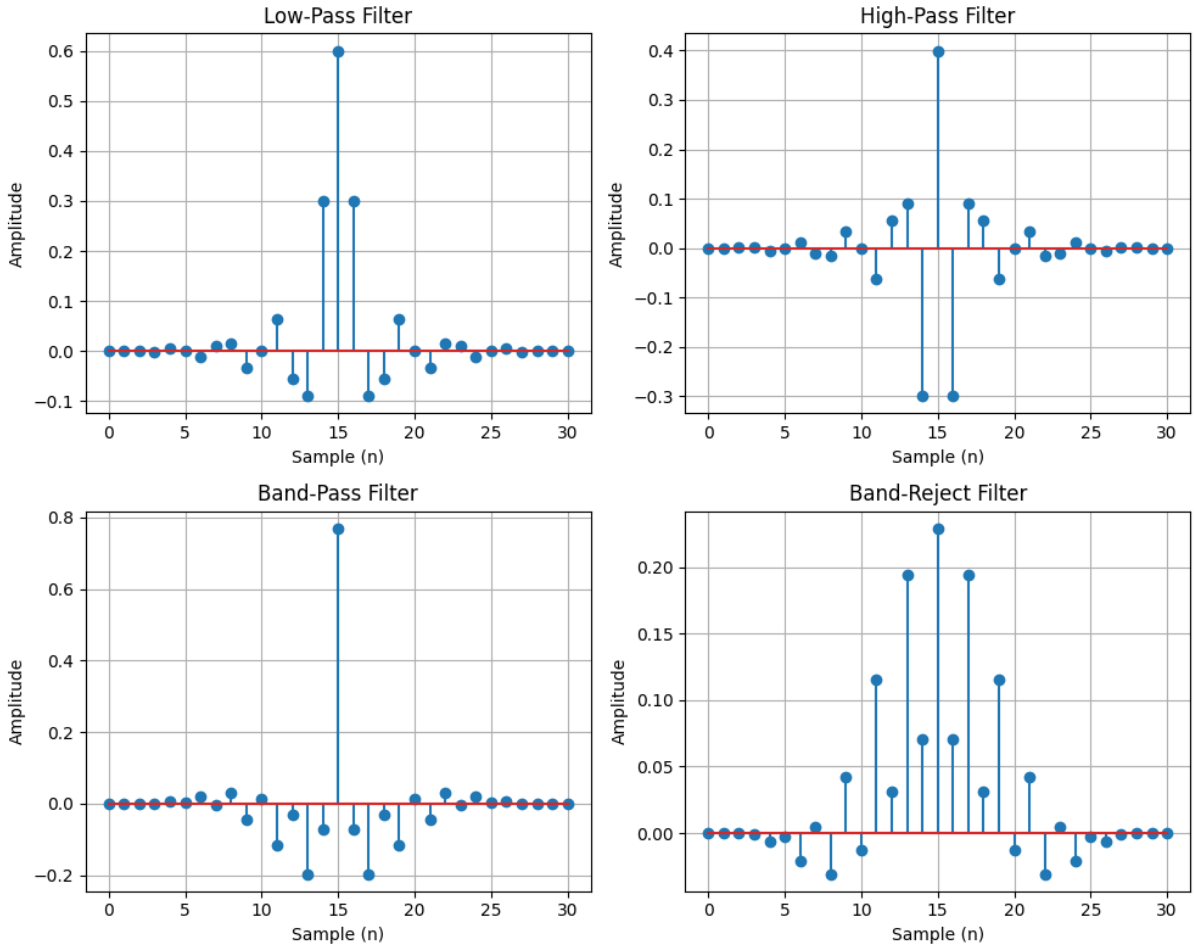
plt.tight_layout()
plt.show()
```

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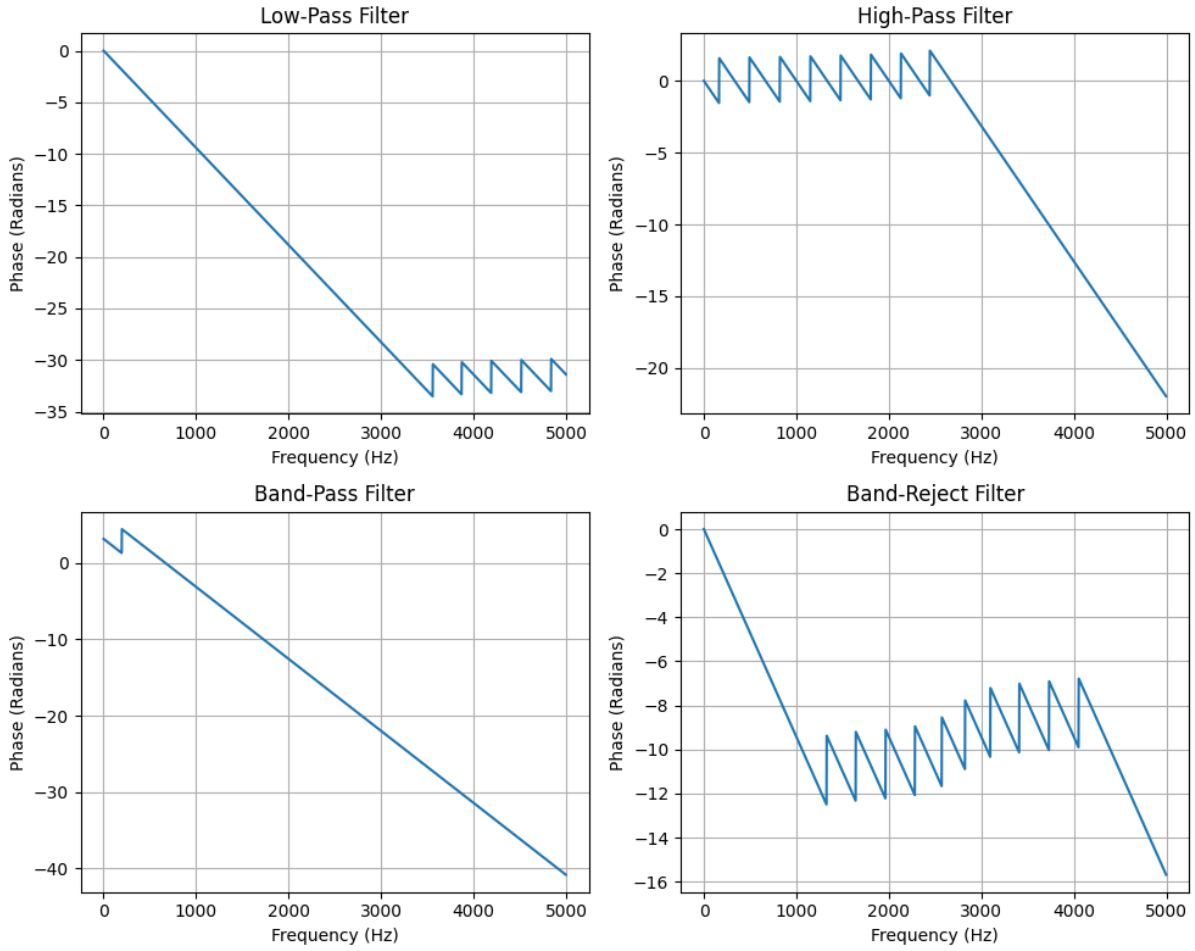
Output :



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Using One Song plot their filtered output.

Code :

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.io import wavfile
from scipy.signal import firwin, freqz, lfilter


# Step 1: Load the song
Fs, song = wavfile.read("/content/Ehasaas.wav")


# Convert stereo song to mono
if song.ndim == 2:
    song = song.mean(axis=1)


# Normalize song
song = song.astype(float)
song = song / (np.max(np.abs(song)) + 1e-10)


print("Sampling Frequency:", Fs)
print("Song Duration:", len(song) / Fs, "seconds")


# Step 2: Filter parameters
N = 31


# 6 kHz requires a sampling frequency greater than 12 kHz
if Fs <= 12000:
    raise ValueError(
        "Use a song with sampling frequency greater than 12000 Hz."
    )


# Step 3: Design four FIR filters
```

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```
low = firwin(N, 3000, window="hann",
             pass_zero="lowpass", fs=Fs)

high = firwin(N, 3000, window="hann",
              pass_zero="highpass", fs=Fs)

band = firwin(N, [3000, 6000], window="hann",
              pass_zero="bandpass", fs=Fs)

stop = firwin(N, [3000, 6000], window="hann",
              pass_zero="bandstop", fs=Fs)

filters = [low, high, band, stop]

titles = ["Low-Pass Filter", "High-Pass Filter",
          "Band-Pass Filter", "Band-Stop Filter"]

# Step 4: Plot magnitude, impulse and phase responses
plt.figure(figsize=(12, 12))

for i in range(4):
    b = filters[i]

    f, H = freqz(b, worN=2048, fs=Fs)

    # Magnitude response
    magnitude = 20 * np.log10(np.maximum(np.abs(H), 1e-10))

    # Impulse response
    impulse = b

    # Phase response
```

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```
    phase = np.unwrap(np.angle(H))

    # Plot magnitude
    plt.subplot(4, 3, i * 3 + 1)
    plt.plot(f, magnitude)
    plt.title(titles[i] + " - Magnitude")
    plt.xlabel("Frequency (Hz)")
    plt.ylabel("Magnitude (dB)")
    plt.grid()

    # Plot impulse
    plt.subplot(4, 3, i * 3 + 2)
    plt.stem(np.arange(N), impulse)
    plt.title(titles[i] + " - Impulse")
    plt.xlabel("Sample (n)")
    plt.ylabel("Amplitude")
    plt.grid()

    # Plot phase
    plt.subplot(4, 3, i * 3 + 3)
    plt.plot(f, phase)
    plt.title(titles[i] + " - Phase")
    plt.xlabel("Frequency (Hz)")
    plt.ylabel("Phase (Radians)")
    plt.grid()

plt.tight_layout()
plt.show()

# Step 5: Apply filters to the song
plt.figure(figsize=(12, 8))
```

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```
for i in range(4):
    filtered_song = lfilter(filters[i], 1, song)

    # Display filtered song waveform
    plt.subplot(4, 1, i + 1)
    time = np.arange(len(song)) / Fs
    plt.plot(time, filtered_song)
    plt.title(titles[i] + " - Filtered Song")
    plt.xlabel("Time (seconds)")
    plt.ylabel("Amplitude")
    plt.grid()

    # Save filtered song
    output = np.int16(
        filtered_song / (np.max(np.abs(filtered_song)) + 1e-10)
        * 32767
    )
    wavfile.write(
        ["low_pass.wav", "high_pass.wav",
         "band_pass.wav", "band_stop.wav"][i],
        Fs, output
    )

plt.tight_layout()
plt.show()

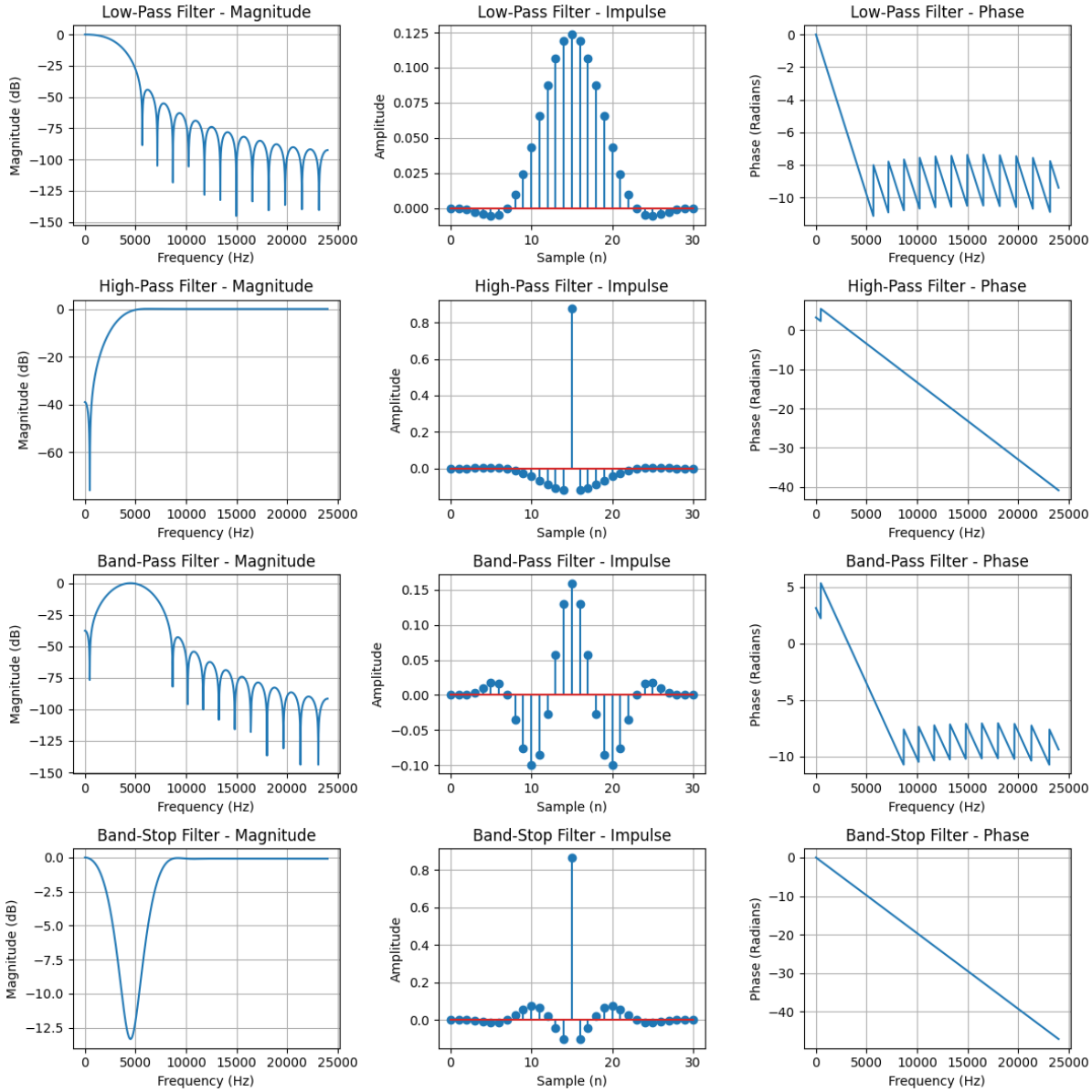
print("All four filtered songs have been saved!")
```

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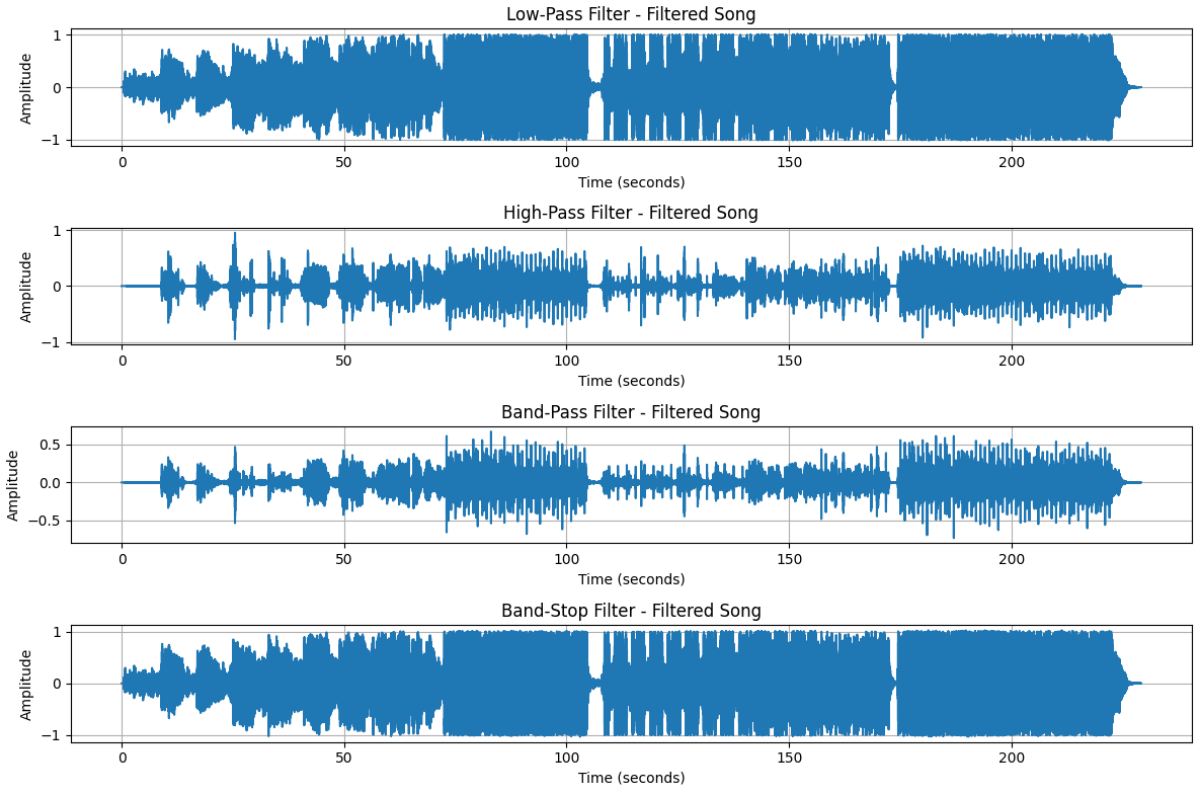
Output :

Sampling Frequency: 48000

Song Duration: 229.09297916666668 seconds



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Conclusion :

In this experiment FIR low-pass, high-pass, band-pass, and band-stop filters using the windowing method (Hann window) with N=31. Magnitude, impulse, and phase responses were plotted and verified for the given cutoff frequencies. The filters were also applied to a song to observe practical filtering effects, and the outputs were saved as WAV files.